

Yusuf Ali

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EDUCATION

Texas A&M University

May 2027

Bachelor of Science in Computer Science; Neuroscience Minor

College Station, TX

- **Involvements:** Aggies Create (Team Lead), Student Engineers' Council (EnVision Conference Chair)
- **Coursework:** Program Design, Data Structures & Algorithms, Discrete Structures, Computer Organization

EXPERIENCE

Neuroengineer Intern

May 2025 – Present

Paradromics

Austin, TX

- Optimizing neural data transmission and machine-learning models to enhance decoding accuracy of the Connexus® Brain-Computer Interface.
- Building a downstream application to showcase the data rate capabilities of the BCI platform.

Curriculum Developer & Instructor

March 2022 – Present

LearnToBot

The Woodlands, TX

- Teaching over 100 students how to build and code problem-solving robots in weekly camps.
- Designing and testing software/hardware projects for future camps and classes.
- Hosting robotics camps for *Destination Imagination* teams in our community.

Robotic Surgery Intern

June 2022 – July 2022

Laredo Sports Medicine Clinic

Laredo, TX

- Developed machine learning algorithms for *Stryker Mako* surgical machines and worked with departments in the clinic including research and development, physical training, and marketing.
- Met with professionals involved in the biotechnology industry from Texas A&M International University.

PROJECTS

Brain-Controlled E-Wheelchair | *EMOTIV, OpenBCI, Python, Arduino, Soldering*

August 2024 – Present

Objective: Develop a brain-computer interface system to control an e-wheelchair using thoughts as commands.

- Disassembled an e-wheelchair controller and soldered its test points to an Arduino Nano.
- Trained mental commands using an EMOTIV headset and converted them to inputs for movement.
- Presented our prototype at the Aggies Create Innovation Expo and placed 1st.

Nixie Tube Watch | *C++, C, PCB Design, Soldering*

June 2023 – January 2024

Objective: Build an analog watch from scratch using cathode ray Nixie tubes from the 1950s.

- Designed printed circuit boards to keep the form factor of the watch as compact as possible.
- Created a Qi standard charging module for the watch to charge wirelessly.
- Built the watch case using resin and CNC-cut glass.
- Programmed the time functions of the watch in C.

TECHNICAL SKILLS

- **Languages:** Python, Java, C, C++, C#, HTML, JavaScript
- **Development Tools:** VS Code, PyCharm, AutoCAD, KiCad, Blender, Ableton
- **Libraries:** Pandas, BeautifulSoup, NumPy, Matplotlib, TensorFlow, scikit-learn
- **Awards:** Aggies Create Innovation Expo Winner, Meloy Engineering Innovators Grant, UT Austin CS Robotics Champion